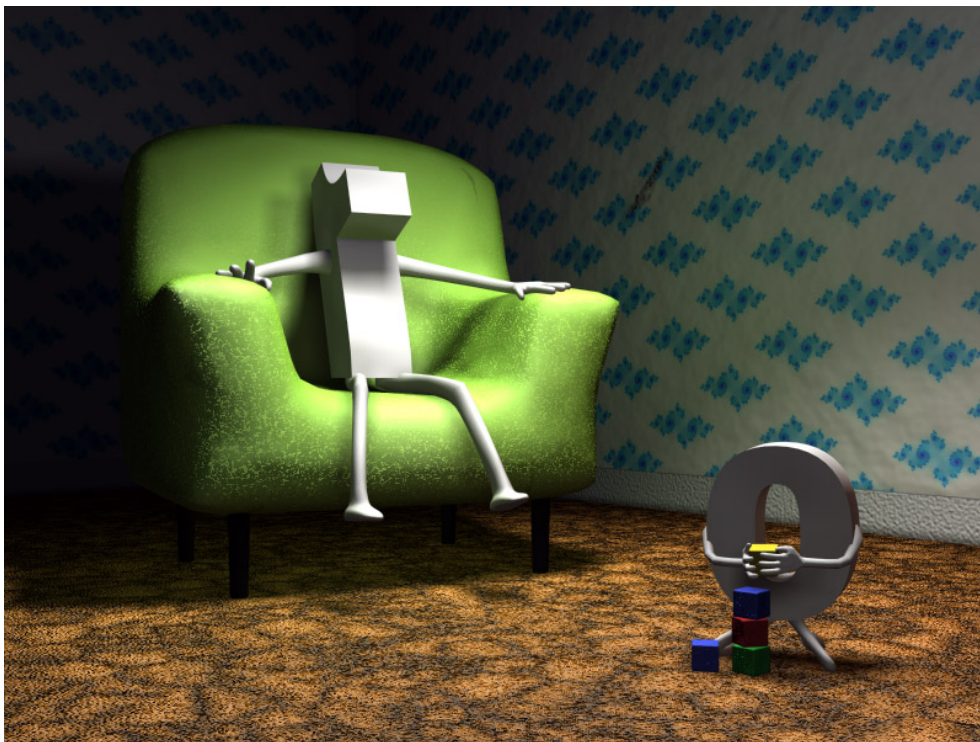


Learning how to prove Workshop Week 3

Integers



Solving equations in $\mathbb{Z}_n$. The Chinese Remainder Theorem

1. Let p be a prime.

(a) Prove that the only elements in $\mathbb{Z}_p$ that are self-inverses are $\bar{1}$ and $\overline{p-1}$.

(b) **Prove Wilson's Theorem:** $(p-1)! \equiv -1 \pmod{p}$

Solution: (a) Notice that $\overline{p-1} = \overline{-1}$, so it is clear that $\bar{1}$ and $\overline{p-1}$ are self-inverses. Suppose that $\bar{a} \in \mathbb{Z}_p$ satisfies $(\bar{a})^2 = \bar{1}$, and let us show that either $\bar{a} = \bar{1}$ or $\bar{a} = \overline{p-1}$.

From $(\bar{a})^2 = \bar{1}$ we obtain that $(\bar{a})^2 - \bar{1} = \bar{0}$. That is to say that $(\bar{a} + \bar{1})(\bar{a} - \bar{1}) = \bar{0}$. Now, $\bar{a} + \bar{1} = \overline{a+1}$ and $\bar{a} - \bar{1} = \overline{a-1}$. So, we have that

$$\overline{a+1} \cdot \overline{a-1} = \bar{0}.$$

We proved in class that in $\mathbb{Z}_p$, if $\bar{b} \cdot \bar{c} = \bar{0}$, then either $\bar{b} = \bar{0}$ or $\bar{c} = \bar{0}$. Using this we obtain that either $\overline{a+1} = \bar{0}$ or $\overline{a-1} = \bar{0}$; if $\overline{a+1} = \bar{0}$, then $\bar{a} = \overline{-1} = \overline{p-1}$; if $\overline{a-1} = \bar{0}$, then $\bar{a} = \bar{1}$.

Alternatively, if you don't remember the theorem we proved in class, you can reason like this:

From $(\bar{a})^2 = \bar{1}$ we obtain that $\overline{a^2 - 1} = \bar{0}$, which means that $p|(a^2 - 1)$. But $a^2 - 1 = (a+1)(a-1)$, so $p|(a+1)(a-1)$. Using now that p is prime we have that either $p|(a+1)$ or $p|(a-1)$.

- If $p|(a+1)$, then $\overline{a+1} = \bar{0}$, that is, $\bar{a} = \overline{-1} = \overline{p-1}$.
- If $p|(a-1)$, then $\overline{a-1} = \bar{0}$, that is, $\bar{a} = \bar{1}$.

(b) Let $\mathbb{Z}_p = \{\bar{0}, \bar{1}, \bar{2}, \dots, \overline{p-1}\}$. We have proved in class that every nonzero element $\bar{a} \in \mathbb{Z}_p$ has an inverse. On the other hand, we know from (a) that $\bar{1}$ and $\overline{p-1}$ are the only self-inverse elements. Thus, grouping in pair of inverses in the product

$$\bar{2} \cdot \bar{3} \cdots \overline{p-2},$$

we get that

$$\bar{2} \cdot \bar{3} \cdots \overline{p-2} = \bar{1}.$$

Next, multiplying both sides of the equation by $\overline{p-1}$ we obtain that

$$\bar{2} \cdot \bar{3} \cdots \overline{p-2} \cdot \overline{p-1} = \overline{p-1} = \overline{-1}, \quad \text{in } \mathbb{Z}_p.$$

Observe that $\bar{2} \cdot \bar{3} \cdots \overline{p-2} \cdot \overline{p-1} = \overline{2 \cdot 3 \cdots (p-2) \cdot (p-1)} = \overline{(p-1)!}$. Thus:

$$\overline{(p-1)!} = \overline{-1}, \quad \text{in } \mathbb{Z}_p \Leftrightarrow (p-1)! \equiv -1 \pmod{p},$$

as desired.

2. Use the Chinese Remainder Theorem to solve the following simultaneous congruence:

$$3x \equiv 2 \pmod{5} \quad \text{and} \quad 2x \equiv 1 \pmod{3}.$$

Solution:

First of all notice that we can apply the Classical Chinese Remainder Theorem since $\gcd(3, 5) = 1$. There are two ways to approach this problem: either we simplify both equations to have them like in the statement of the theorem, or we simplify only one of them and proceed like in the proof of the Classical Chinese Remainder Theorem. I include both options below.

Here, we are going to use that the inverse of $\overline{p-1}$ in $\mathbb{Z}_p$ is itself; that is:

$$\overline{p-1}^{-1} = \overline{p-1} \text{ in } \mathbb{Z}_p.$$

For $p = 3$, we get that the inverse of $\bar{2}$ in $\mathbb{Z}_3$ is $\bar{2}$; multiplying the second congruence by 2, we obtain $x \equiv 2 \pmod{3}$. Therefore, we have to solve the simultaneous congruences:

$$3x \equiv 2 \pmod{5} \quad \text{and} \quad x \equiv 2 \pmod{3}$$

The solutions of $x \equiv 2 \pmod{3}$ are $x = 2 + 3k$, with $k \in \mathbb{Z}$. Substituting in the first congruence gives:

$$3(2 + 3k) \equiv 2 \pmod{5} \Leftrightarrow 9k \equiv -4 \pmod{5}$$

Next, since $9 \equiv 4 \pmod{5}$ and $-4 \equiv 1 \pmod{5}$ we have to solve the following congruence:

$$4k \equiv 1 \pmod{5}$$

Notice that $\bar{4}^{-1} = \bar{4}$ in $\mathbb{Z}_5$ (here, $p = 5$). Using this we get that $k \equiv 4 \pmod{5}$. Thus, taking $k = 4$ we obtain that a common solution of the simultaneous congruences is $x = 2 + 3 \cdot 4 = 14$. By the Chinese Theorem Remainder we can conclude that all the solutions are:

$$14 + (3 \cdot 5)n = 14 + 15n, \quad n \in \mathbb{Z}.$$

Let us now simplify both equations first: the inverse of $\bar{3}$ in $\mathbb{Z}_5$ is $\bar{2}$, while the inverse of $\bar{2}$ in $\mathbb{Z}_3$ is $\bar{2}$. Using these, we obtain the equivalence congruences:

$$x \equiv 4 \pmod{5}, \quad x \equiv 2 \pmod{3}.$$

The solutions of $x \equiv 4 \pmod{5}$ are of the form $x = 4 + 5k$, for some $k \in \mathbb{Z}$. Substituting in the second congruence we obtain:

$$4 + 5k \equiv 2 \pmod{3} \Rightarrow 5k \equiv -2 \equiv 1 \pmod{3}.$$

Now the inverse of $\bar{5}$ in $\mathbb{Z}_3$ is $\bar{2}$; so multiplying by $\bar{2}$ the previous equation we obtain:

$$k \equiv 2 \pmod{3}.$$

Thus, a common solution of the given congruences is $x = 4 + 5 \cdot 2 = 14$. The Chinese Remainder Theorem tells us that any other solution is congruent to 14 modulo $3 \cdot 5 = 15$. In other words, all the common solutions are of the form $14 + 15n$, for $n \in \mathbb{Z}$.